



**Waiting for everyone
to join**

**TU257
Information Systems
Marisa Llorens**

Week 5 Information Systems

TU257

Agenda

- Normalisation
 - UNF
 - 1NF
 - 2NF
 - 3NF

Normalisation

- The process of designing a database in the **most efficient** way possible.
 - A technique for designing relational database tables to minimise **duplication of information** and, in doing so, to safeguard the database against certain types of **logical inconsistency**.
- More highly normalised tables greatly **simplify development** and **maintenance** of the database and **contribute** to its **extensibility**.

Normalisation

- Properly executed, the normalisation process **eliminates uncontrolled data redundancies**,
 - thus **eliminating** the **data anomalies**, and
 - the **data integrity problems** that are produced by such redundancies.

Stages of Normalisation

- **Unnormalised**

 - ↓ Remove repeating groups

- **1NF**

 - ↓ Remove partial dependencies

- **2NF**

 - ↓ Remove transitive dependencies

- **3NF**

Repeating Groups

- A **repeating group** is an attribute (or set of attributes) that can have **more than one value** for a **primary key value**.
- **Repeating Groups** are **not allowed** in a relational design, since all attributes have to be '**atomic**' - i.e. there can only be one value per cell in a table.

Repeating Groups

- **Example:** The following relation contains staff and department details and a list of telephone contact numbers for each member of staff.

StaffNo	Job	Dept	DeptName	City	Contact Number
SL10	Salesman	10	Sales	Sligo	018111777, 018111888, 079311122
SA51	Manager	20	Accounts	Dublin	017111777
DS40	Clerk	20	Accounts	Dublin	
OS45	Clerk	30	Operations	Dublin	079311555

Repeating Groups

Invoice Number: 1234

Client ID: 432

Date:

01/01/2023

Product ID	Product Name	Quantity	Price	Total
43	Table	1	€ 900.00	€ 900.00
23	Chair	6	€ 190.00	€ 1,140.00
455	Pillow	4	€ 42.00	€ 168.00
23	Cushion	4	€ 25.00	€ 100.00
Total				€ 2,308.00

Repeating Groups

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23	Cushion	4	€ 25.00	€ 100.00
Total				€ 2,308.00

Invoice Number	ClientID	OrderDate	ProductID	Product Name	Quantity	Price	total
1234	432	2023-01-01	43,23,455,23	Table, Chair, Pillow, Cushion	1,6,4,4	900,190,42,25	900,1140,168,100

Dependencies

- **Functional Dependency:**
 - **Fully Functional Dependency**
 - **Partial Functional Dependency**
- **Transitive Dependency**

Functional Dependency

- **Formal Definition:** Attribute **B** is functionally dependent upon attribute **A** (or a collection of attributes) if a value of **A** determines a single value of attribute **B** at any one time.
- In other words, attribute **B** has a functional dependency on attribute **A** if, for each value of **A**, there is exactly one value of attribute **B**.

Functional Dependency

Formal Notation $A \rightarrow B$

- This should be read as '**A determines B**' or '**B is functionally dependent on A**'. **A** is called the determinant and **B** is called the object of the determinant.

StaffNo	Job	Dept	DeptName
SL10	Salesman	10	Sales
SA51	Manager	20	Accounts
DS40	Clerk	20	Accounts
OS45	Clerk	30	Operations

Functional Dependency Example

Dept → **DeptName**

This should be read as '**Department determines DepartmentName**'
or

'DepartmentName **is functionally dependant on Department**'.



StaffNo	Job	Dept	DeptName
SL10	Salesman	10	Sales
SA51	Manager	20	Accounts
DS40	Clerk	20	Accounts
OS45	Clerk	30	Operations

Dept → **DeptName**

Full Functional Dependency

- An attribute is fully functional dependent on a set of attributes **X** if it is...
 - functionally dependent on **X**, and
 - not functionally dependent on any proper subset of **X**
 - **Only relevant for composite determinants**

Full Functional Dependency

Only of relevance with composite determinants.

All the attributes of the composite determinant (Course# and Subject_Code) are needed to determine the Subject Type.

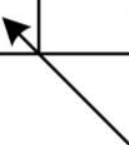
Course#	Subject_Code	Subject_Type
ICA40299	ICAITB060A	Core
ICA40299	ICAITU018B	Elective
ICA40199	ICAITU018B	Core

Partial Functional Dependency

- This is the situation that exists if it is necessary to only use a **subset** of the attributes of the composite determinant to identify its object uniquely.

StudentNo	UnitNo	Room	Grade
D1000100	A01	TH224	2
D1000010	A01	TH224	14
D1001011	A02	JS075	3
D1000001	A01	TH224	16

Repetition
of data



Full Functional Dependencies

$(\text{StudentNo}, \text{UnitNo}) \rightarrow \text{Grade}$

Partial Functional Dependencies

$\text{UnitNo} \rightarrow \text{room}$

Transitive Dependency

- A **transitive dependency** exists when there is an intermediate functional dependency.
- A condition in which an attribute is dependent on another attribute that is not part of the primary key.
 - i.e. an indirect functional dependency,
 - one in which $X \rightarrow Z$ only by virtue of $X \rightarrow Y$ and $Y \rightarrow Z$

Transitive Dependency

Formal Notation

- If $A \rightarrow B$ and $B \rightarrow C$, then it can be stated that the following transitive dependency exists:

$$A \rightarrow B \rightarrow C$$

Transitive Dependency

Example:

Company	CEO	age
Microsoft	Satya Nadella	56
Alphabet/ Google	Sundar Pichai	51
Alibaba	Jack Ma	59

If **company** $\rightarrow$ **CEO** and **CEO** $\rightarrow$ **age**, then it can be stated that the following transitive dependency exists:

company $\rightarrow$ **CEO** $\rightarrow$ **age**

Normal Forms

When is a table in 1NF?

- A table is in **1NF** when...
 - All the key attributes are defined (**no repeating groups** in the table) and when,
 - All remaining attributes are **dependent** on the primary key
 - **Note:** A table in **1NF** still **may contain partial dependencies**, i.e., dependencies based on **only part** of the primary key.

Normal Forms

When is a table in 2NF?

- A table is in **2NF** when
 - it is in **1NF** and,
 - it includes **no partial dependencies**.

Note: A table in **2NF** still may contain **transitive dependencies**, i.e., dependencies based on attributes that are **not part** of the primary key.

Normal Forms

When is a table in **3NF**?

- A table is in **3NF** when...
 - it is in **2NF**, and
 - it contains no **transitive dependencies**.

Un-normalised Normal Form (UNF or ONF)

- **Un-normalised Normal Form (UNF or ONF)**
 - **Definition:** A relation is un-normalised when it has not had any normalisation rules applied to it, and it suffers from various anomalies.
 - This only tends to occur where the relation has been designed using a '*bottom-up approach*' i.e. the capturing of attributes to a 'universal relation' from a screen layout, manual report, manual document, etc...

Example: Logical Design ('Bottom-up' Approach)

ORDER

Customer No: 123456

Order Number: 00654321

Name: Joe Bloggs

Order Date: 10-Oct-2011

Address: 1 Bolton Street

Dublin

NP6-30-DF7

Product Number	Product Description	Unit Price (€)	Order Quantity	Line Total (€)
T5060	Hook	5.00	5	25.00
PT42	Bolt	2.50	10	20.50
QZE48	Spanner	20.00	1	20.00

Order Total (incl. VAT):

65.50

Example: Place in UNF

- Gather attributes in the following schema form:
- **Entity**(non-repeating attributes
(repeating attributes)*)

Calculated fields

Repeating attributes

ORDER

Customer No: 123456

Order Number: 00654321

Name: Joe Bloggs

Order Date: 10-Oct-2011

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QZE48	Spanner	20.00	1	20.00

Order Total (incl. VAT): 65.50

Example: Place in UNF

Entity(non-repeating attributes (repeating attributes)*)

- Un-normalised Normal Form (UNF or ONF)

ORDERS

cust_no	cust_name	cust_add	order_no	order_date	prod_no	prod_desc	unit_price	order_qty	line_total	order_total
---------	-----------	----------	----------	------------	---------	-----------	------------	-----------	------------	-------------

Orders(cust_no,cust_name,cust_add, order_no, order_date,(product_no,
prod_desc_unit_price, order_qty, line_total)*, order_total)

First Normal Form (1NF)

Definition: A relation is in **1NF** if, and only if, all its underlying attributes contain 'atomic' values only.

Remove **repeating groups** into a new relation

ORDERS (order_no, order_date, cust_no, cust_name, cust_add,
(prod_no, prod_desc, unit_price, ord_qty, line_total)*, order-total)

Example: UNF to 1NF

1.Remove the outermost **repeating group** (and any nested repeated groups it may contain) and create a new relation to contain it. Rename original to indicate **1NF**.

ORDERS-1 (**order_no**, order_date, cust_no, cust_name, cust_add, **order_total**)

(**prod_no**, **prod_desc**, **unit_price**, **ord_qty**, **line_total**)

Example: UNF to 1NF

2. Add to this relation a copy of the **PK** of the relation immediately enclosing it.

ORDERS-1 (**order_no**, order_date, cust_no, cust_name, cust_add, order_total)

(**order_no**, prod_no, prod_desc, unit_price, ord_qty, line_total)



Example: UNF to 1NF

3. Name the new entity (appending the number **1** to indicate **1NF**).

ORDER-LINES-1 (order_no, prod_no, prod_desc, unit_price, ord_qty, line_total)

4. Determine the **PK** of the new entity.

ORDER-LINES-1 (order_no, prod_no, prod_desc, unit_price, ord_qty, line_total)

Note on choosing PK

- Foreign Keys as part of a composite key can be used as PKs
- A single Foreign key is usually not suitable as Primary Keys as they allow duplication
- Alternatives:
 - Find a field that uniquely identifies each record in the table
 - Add a new field (auto-incrementing integer) to act as the primary key.

Second Normal Form (2NF)

Definition: A relation is in **2NF** if, and only if, it is in **1NF** and every non-key attribute is **fully dependent** on the primary key.

Remove **partial functional dependencies** into a new relation.

Check each relation individually.

Second Normal Form (2NF)

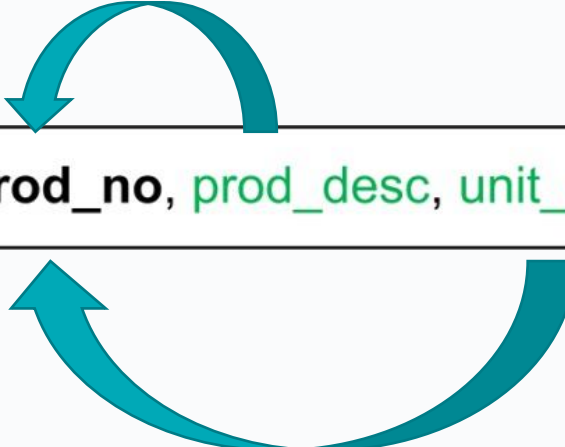
First relation check

ORDERS-1 (**order_no**, order_date, cust_no, cust_name, cust_add, order_total)

Only one single primary key. No partial dependencies.

Second Normal Form (2NF)

Second relation check



ORDER-LINES-1 (order_no, prod_no, prod_desc, unit_price, ord_qty, line_total)

Composite primary key. Partial dependencies found.

Example: 1NF to 2NF

ORDER-LINES-1 (order_no, prod_no, prod_desc, unit_price, ord_qty, line_total)

- 1. Remove** the offending attributes that are only **partially functionally dependent** on the composite key, and place them in a new relation.

ORDER-LINES-1 (order_no, prod_no, ord_qty, line_total)

(prod_desc, unit_price)

Example: 1NF to 2NF

ORDER-LINES-1 (order_no, prod_no, prod_desc, unit_price, ord_qty, line_total)

2.Add to this relation a copy of the attribute(s) which determines these offending attributes. These will automatically become the primary key of this new relation...

ORDER-LINES-1 (order_no, prod_no, ord_qty, line_total)

(prod_no, prod_desc, unit_price)



Example: 1NF to 2NF

3. Name the new entity (appending the number 2 to indicate 2NF).

PRODUCTS-2 (prod_no, prod_desc, unit_price)

4. Rename the original entity (ending with a 2 to indicate 2NF).

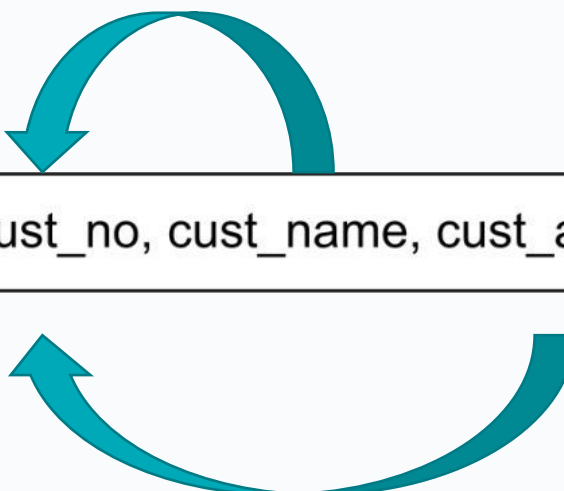
ORDER-LINES-2 (order_no, prod_no, ord_qty, line_total)

Third Normal Form (3NF)

Definition: A relation is in **3NF** if, and only if, it is in **2NF** and every non-key attribute is **non-transitively dependent** on the primary key.

Remove **transitive dependencies** into a new relation

Third Normal Form (3NF)



ORDERS-2 (order_no, order_date, cust_no, cust_name, cust_add, order_total)

Cust_no is not part of the primary key

Example: 2NF to 3NF

ORDERS-2 (**order_no**, order_date, cust_no, cust_name, cust_add, order_total)

1.Remove the offending attributes that are **transitively dependent** on non-key attributes, and place them in a new relation.

ORDERS-2 (**order_no**, order_date, cust_no, order_total)

(cust_name, cust_add)

Example: 2NF to 3NF

ORDERS-2 (order_no, order_date, cust_no, cust_name, cust_add, order_total)

2. Add to this relation a copy of the attribute(s) which determines these offending attributes. These will automatically become the primary key of this new relation...

ORDERS-2 (order_no, order_date, cust_no, order_total)

(cust_no, cust_name, cust_add)



Example: 2NF to 3NF

3.Name the new entity (appending the number 3 to indicate 3NF).

CUSTOMERS-3 (cust_no, cust_name, cust_add)

4.Rename the original entity (ending with a 3 to indicate 3NF).

ORDERS-3 (order_no, order_date, cust_no, order_total)

Example: Relations in Highest Normal Form

- Logical Schema:

ORDERS-3 (order_no, order_date, cust_no)

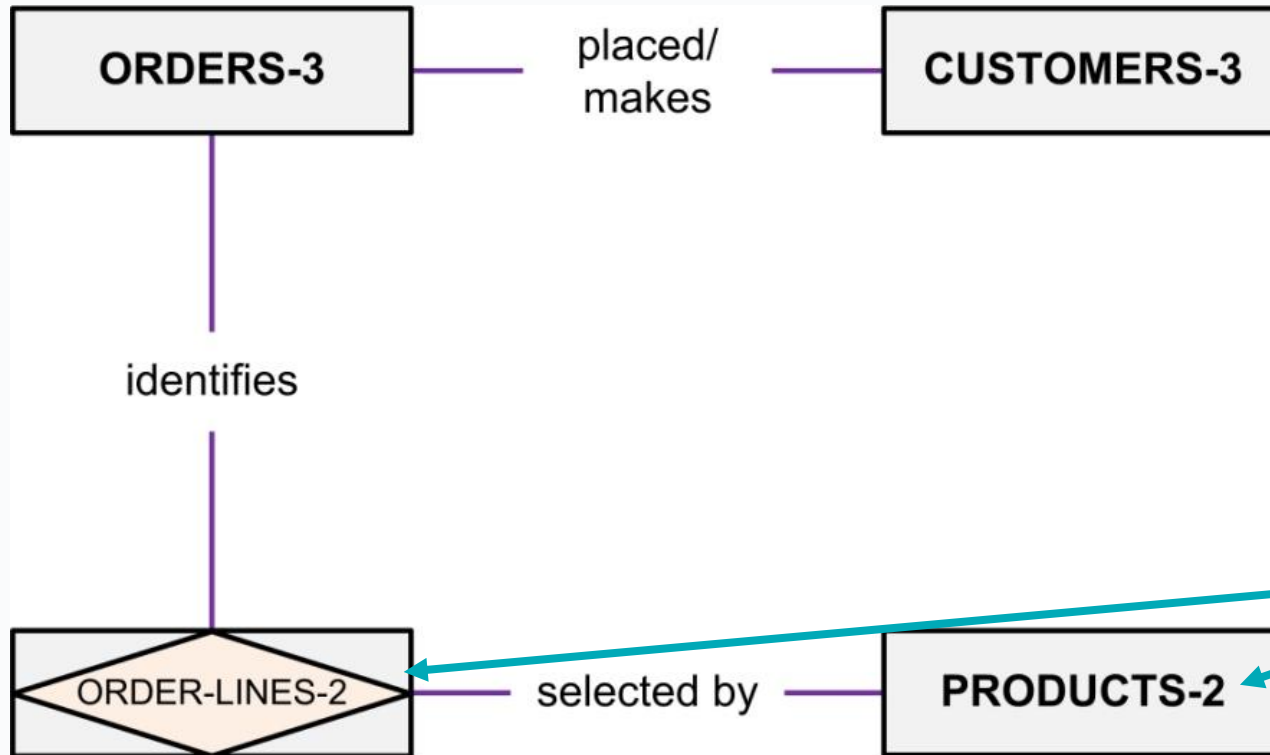
CUSTOMERS-3 (cust_no, cust_name, cust_add)

PRODUCTS-3 (prod_no, prod_desc, unit_price)

ORDER-LINES-3 (order_no, prod_no, ord_qty)

Note! Calculated fields are removed

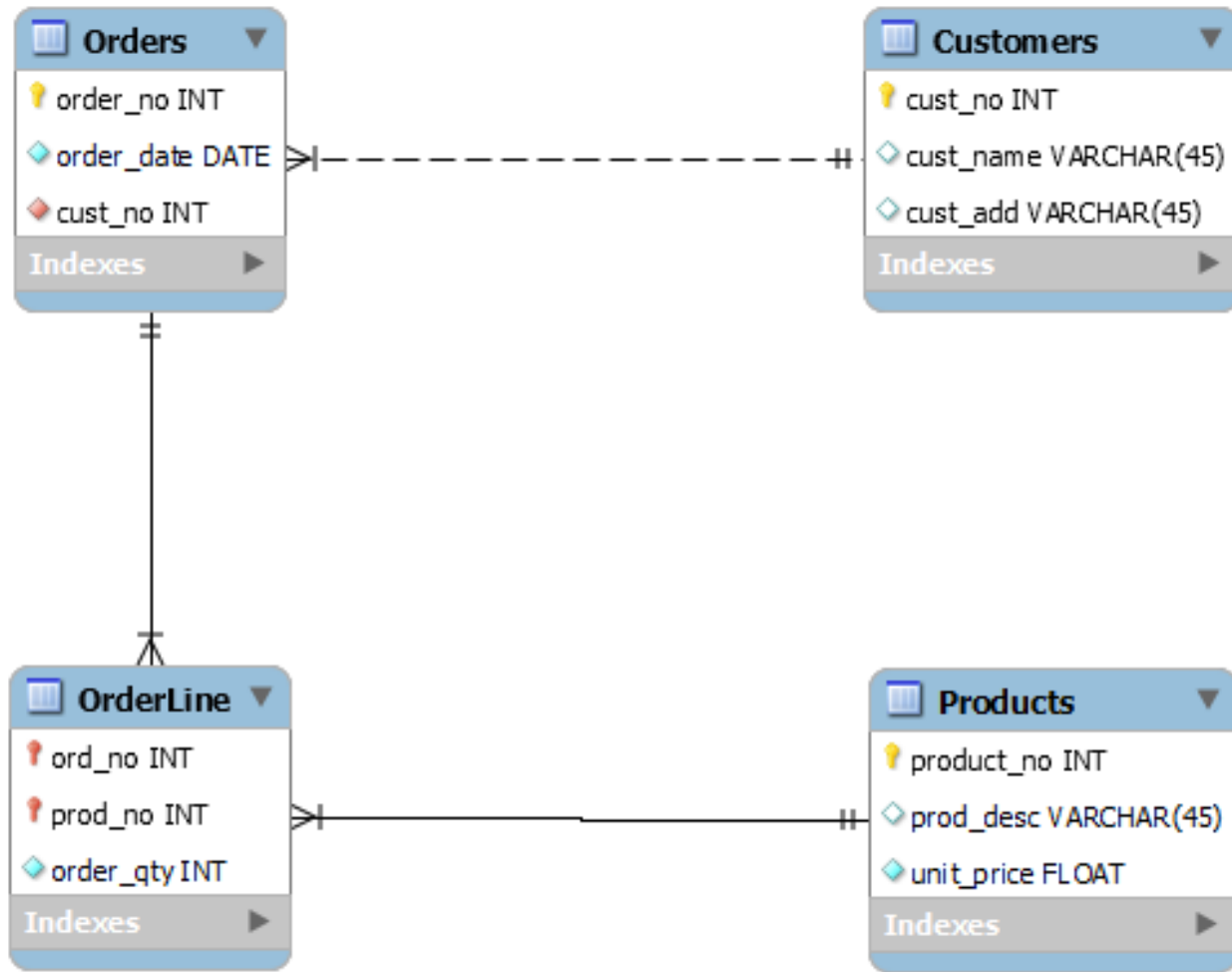
Example: E-R Diagrams



ORDERS-3 (order_no, order_date, cust_no)
CUSTOMERS-3 (cust_no, cust_name, cust_add)
PRODUCTS-3 (prod_no, prod_desc, unit_price)
ORDER-LINES-3 (order_no, prod_no, ord_qty)

Also in 3-NF

Example: E-R Diagrams



ORDERS-3 (order_no, order_date, cust_no)
CUSTOMERS-3 (cust_no, cust_name, cust_add)
PRODUCTS-3 (prod_no, prod_desc, unit_price)
ORDER-LINES-3 (order_no, prod_no, ord_qty)

**Fully-Attributed Key-Based E-R
Diagram**

Normalisation Example

PPS number	Empl_Name	Empl_Role	Empl_salary	Empl_email
123	Mary Smith	Manager	4000	mary@tup.ie , m.white@tup.ie , whitem@tup.ie
432	John Murphy	Admin	2000	John.murphy@tup.i e
524	Adele Ryan	Team leader	2500	a.ryan@tup.ie , ryana@tup.ie
8828	Sonya Kennedy	Admin	2000	s.kennedy@tup.ie

Normalisation Example

PPS number	Empl_Name	Empl_Role	Empl_salary	Empl_email
123	Mary Smith	Manager	4000	mary@tup.ie , m.white@tup.ie , whitem@tup.ie
432	John Murphy	Admin	2000	John.murphy@tup.i e
524	Adele Ryan	Team leader	2500	a.ryan@tup.ie , ryana@tup.ie
8828	Sonya Kennedy	Admin	2000	s.kennedy@tup.ie

- Salary depends on role
- PK PPS number

Example Company

PPS number	Empl_Name	Empl_Role	Empl_salary	Empl_email
123	Mary Smith	Manager	4000	mary@tup.ie , m.white@tup.ie , whitem@tup.ie
432	John Murphy	Admin	2000	John.murphy@tup.i e
524	Adele Ryan	Team leader	2500	a.ryan@tup.ie , ryana@tup.ie
8828	Sonya Kennedy	Admin	2000	s.kennedy@tup.ie

- What normal form is this table in?



Short break

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Information Systems

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Plan for next few weeks

- Week 6
 - Revision class
 - Lab SQLite 1
- Week 7
 - Quiz
 - No Lab
- Week 8
 - Submission of Assignment
 - Lab SQLite 2

Thanks

- Edmund Nevin for a earlier versions of these lecture notes



**Break
Back at 8pm
for Lab session**

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